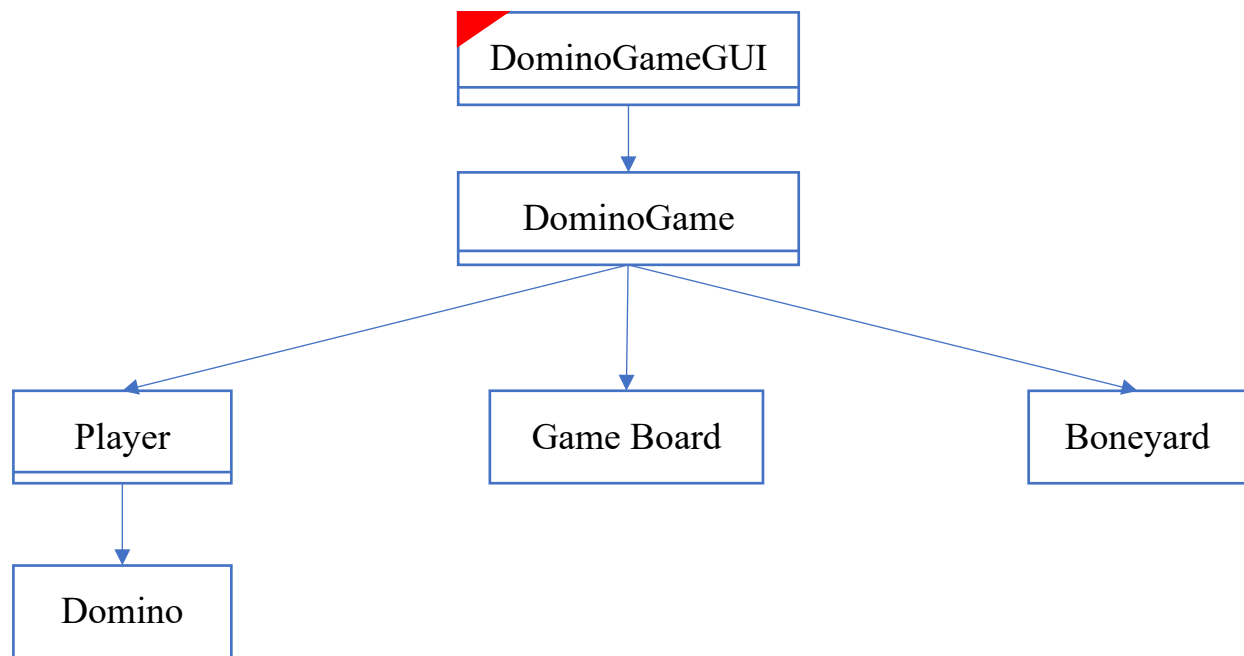


Proposed Design of Version 2



Trigger: update from mouse and keyboard

Description of the Design

DominoGameGUI : Manages the graphical user interface for the Domino game.

- Data:
 - game: The main game logic object
 - boardPanel: Panel for displaying the game board
 - playerHandPanel: Panel for displaying the player's hand
 - drawButton: Button for drawing dominos
 - messageLabel: Label for displaying game messages
- Behavior:
 - initializeUI(): Initialize the user interface components
 - drawBoard(Graphics): Draws the game board with the current state of played dominos.
 - drawDomino(Graphics, Domino, int, int): Draw a single domino on the board
 - updateUI(): Updates all UI components to reflect the current game state
 - updatePlayerHand(): Updates the display of the player's hand
 - playDomino(Domino): Handles the logic when a player attempts to play a domino
 - updateMessage(messageLabel): Updates the message displayed to the user.

DominoGame : Implements the core game logic and manages the game flow.

- Data:
 - human: Represents the human player
 - computer: Represents the computer player
 - boneyard: The pool of undrawn dominos
 - board: the current state of the game board
 - lastPlayer: Player who made the last move
 - gameOver: Indicates if the game has ended
- Behavior:
 - initializeGame(): Initializes a new game.
 - dealInitialHands(): Deals the initial hands to both players.
 - drawForHuman(): Allows the player to take a domino from the boneyard.
 - playHumanDomino(Domino, boolean, boolean): Process a human player's move
 - computerTurn(): Manages the computer's turn
 - canPlay(Domino, boolean): Checks if a domino can be played on the board
 - isValidPlay(Domino, boolean): Checks if a play is valid
 - checkGameOver(): Checks if the game is over
 - hasPlayableDomino(Player): Checks if a player has any playable dominos.
 - calculateFinalScore(Player): Calculates the final score for a player.
 - getBoardDominos(): Gets the list of dominos currently on the board.
 - getHumanHand(): Gets the human player's hand
 - canHumanDraw(): Checks if the human player can draw a domino
 - isGameOver(): Checks if the game is over
 - gameStateMessage(): Gets the current game state message
 - resetGame(): Resets the game to initialize the game

GameBoard: Represents the playing area where dominos are placed.

- Data:
 - dominos: List of Domino played on the board
 - dominoPositions: List of positions for each played domino
 - leftEnd: The value on the left end of the board
 - rightEnd: The value on the right end of the board
- Behavior:

- GameBoard(): Initializes an empty game board
- playDomino(Domino, boolean): Put a domino to the board
- getLeftEnd(): Gets the value on the left end of the board
- getRightEnd(): Gets the value on the right end of the board
- getDominos(): Gets a copy of the list of dominos on the board
- getDominoPositions(): Returns the list of domino positions
- isEmpty(): Checks if the board is empty

Player: Represents a player in the game.

- Data:
 - hand: The dominos in the player's hand
 - name: String representing the player's name.
- Behavior:
 - Player(String): Initializes a player with a given name
 - addDomino(Domino): Adds a domino to the player's hand
 - removeDomino(Domino): Removes a domino from the player's hand.
 - getHand(): Gets a copy of the player's hand.

Boneyard: Represents the pool of undrawn dominos.

- Data:
 - dominos: List of dominos in the boneyard
- Behavior:
 - Boneyard(): Initializes the boneyard with a full set of dominos and shuffles them.
 - draw(): Draws a domino from the boneyard.
 - isEmpty(): Checks if the boneyard is empty.

Domino: Represents a single domino piece.

- Data:
 - left: The value of the left side of the domino.
 - right: The value of the left side of the domino.
- Behavior:
 - Domino(int, int): Creates a new domino with given left and right values.
 - getLeft(): Gets the value on the left side of the domino.
 - getRight(): Gets the value on the right side of the domino.
 - rotate(): Rotates the domino, swapping left and right values.
 - getSum(): Calculates the sum of the left and right values.

- toString(): Returns a string representation of the domino.